Disc Golf Disc Types and Stabilities

Timothy Cutts

Executive Summary

This project analyzes disc golf discs based on disc type (putt & approach, mid-range, fairway drivers, and distance drivers) and disc stability (understable, stable, and overstable). Stability rating is based on disc flight numbers of turn minus fade. Any disc that falls below -2 is understable, between -2 and 2 is stable, or over 2 is overstable. The data was collected via web scrapping of the website Trydiscs.com via Python, and the visuals were completed in Tableau. The goal is to identify if some disc types are more geared toward a stability category.

Objective

The objective of this project is to determine if there is a gap in disc golf manufacturing based on disc types and stability. Understanding what the top manufacturers are creating based on disc types or stability helps identify the gaps in their market share. Also, determining if disc types tend to be skewed towards a specific stability shows open opportunities for manufacturers to create newer and different discs.

Research Questions

This project is aimed at answering the following question:

- *Are disc type and stability linked?*

Data Set

The data set used in this project was created by scrapping the website trydiscs.com with a python script. The script queried the website and categorized each disc by their flight numbers, name and manufacturer. It also took the discs turn and fade flight numbers of the discs and determined what stability class it belonged in. The discs were also categorized if they were discontinued or still in production. Any NULL values in flight numbers categorized the discs as an “Unknown” category. The data set was then saved as a comma separated value file.

Data Set Dimensions

This data set contains 2102 rows in the following configuration:

Manufacturer	Disc Name	Speed	Glide	Turn	Fade	Stability	Disc Type	Discontinued

String	String	Integer	Integer	Integer	Integer	Integer	String	Bool

Exploratory Data Analysis

After importing the data set into tableau, we can explore the top 15 manufacturers based on individual disc molds are in descending order: Innova, Discraft, Latitude 64, Prodigy, Discmania, Dynamic Discs, Lone Star Discs, Gateway, MVP, Yikun, Westside, DGA, Axion, Millennium, and Doomsday Discs. The most common disc type is the putt & approach disc, speeds 1 to 4. Innova has almost twice as many putt & approach molds as the next manufacturer, Discraft. Disc stability in order from most to least is: Stable, Overstable, and then Understable. In the Stable and Overstable category the highest quantity of molds is that of the putt & approach type. In the Understable category, fairway drivers are the most produced. Discraft has the most discontinued discs in the unknown category. This could be likely due to them producing prototype discs called “Ace Race 20XX” and then at a later time it becomes a new disc in their line up with a proper name and flight numbers.

Statistical Analysis

For this dataset I performed a Chi-Square test. The Chi-Square test was to see if stability and disc type are independent or connected to each other.

Chi-Square Test:

Null Hypothesis: Stability and disc type are independent of each other.

Alternative Hypothesis: Stability and disc type are not independent of each other.

Variable chosen: Stability and Disc Type

Alpha chosen: 5% (standard)

Results:

Chi-square statistic = 33.41

Degrees of freedom = 6

P-value = 0.00000874

The results from this test show a p-value lower than the chosen alpha level of 5%. This means that we need to reject the null hypothesis and accept the alternative hypothesis. Disc types and stability are related and not independent

of each other. Note that this test does show a relation between disc type and stability but does not determine the strength of that relationship. One disc type may have more affect on the results than others.

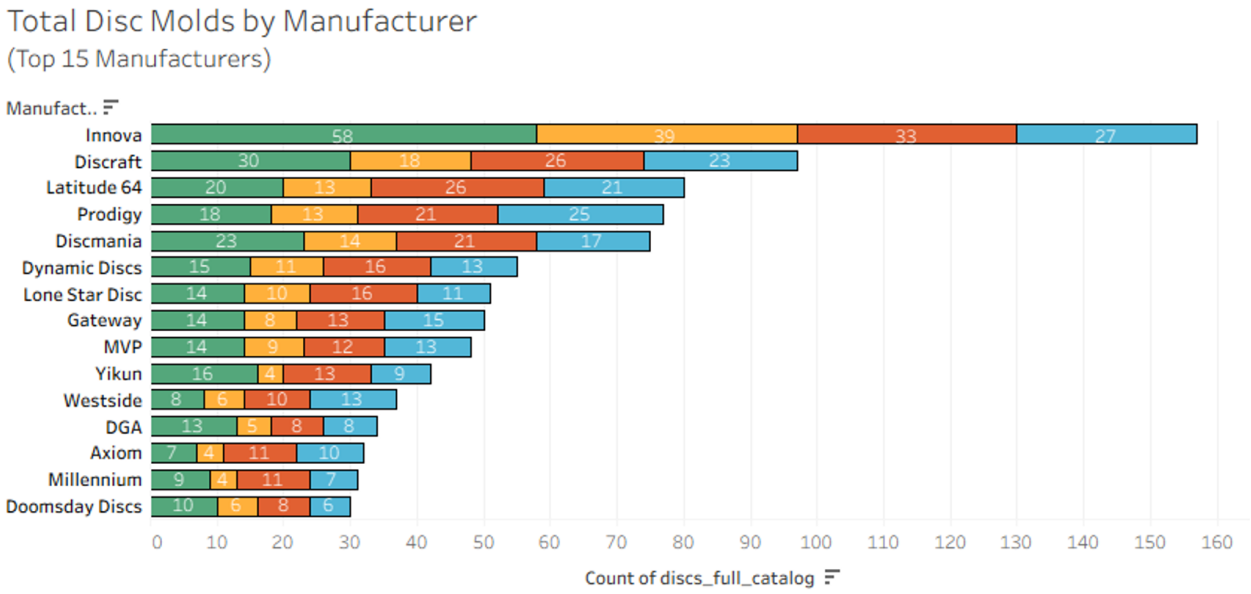
Key Observations

The 15 manufactures by disc mold counts are Innova, Discraft, Latitude 64, Prodigy, Discmania, Dynamic Discs, Lonestar, Gateway, MVP, Yikun, Westside, Axiom, Millennium, and Doomsday Discs.

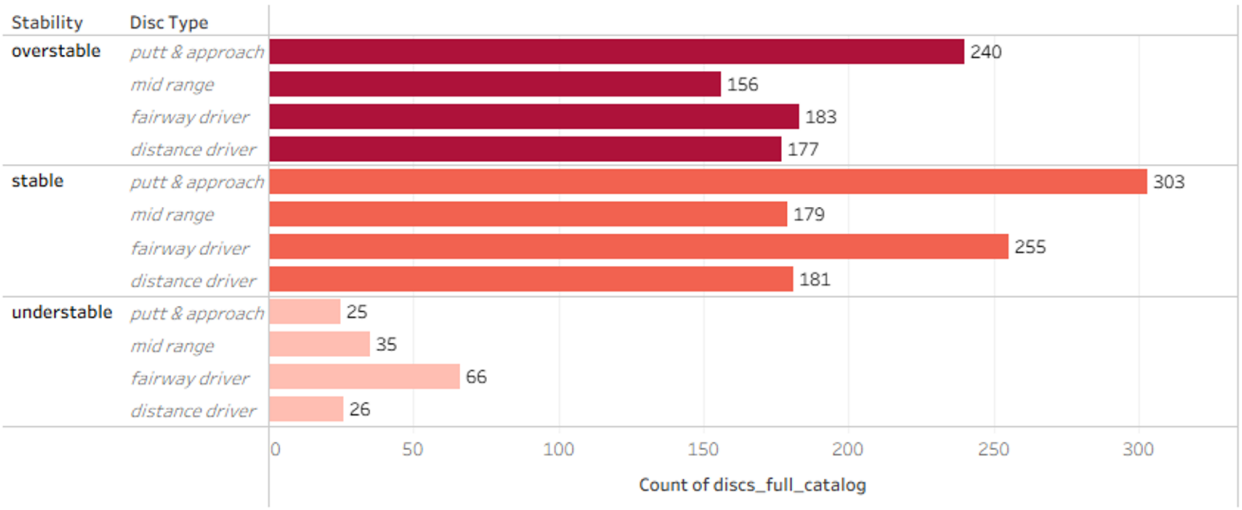
The stability with the most molds is in the stable category, with 918 molds, followed by overstable with 756 and then the understable category comes in at a 152 molds. Every disc type follows this in stability.

Understable disc molds are the least represented in all categories. The reason for this is unclear but it could be due to forehand throws change which direction the disc falls to the ground. There might not be a need for every disc to have a counter part in the opposite direction.

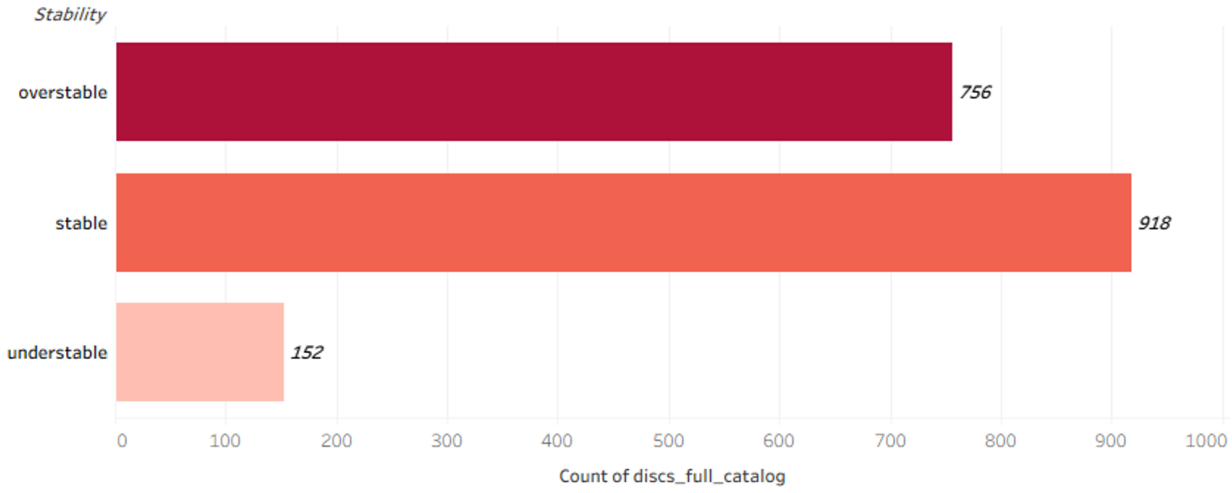
Visuals

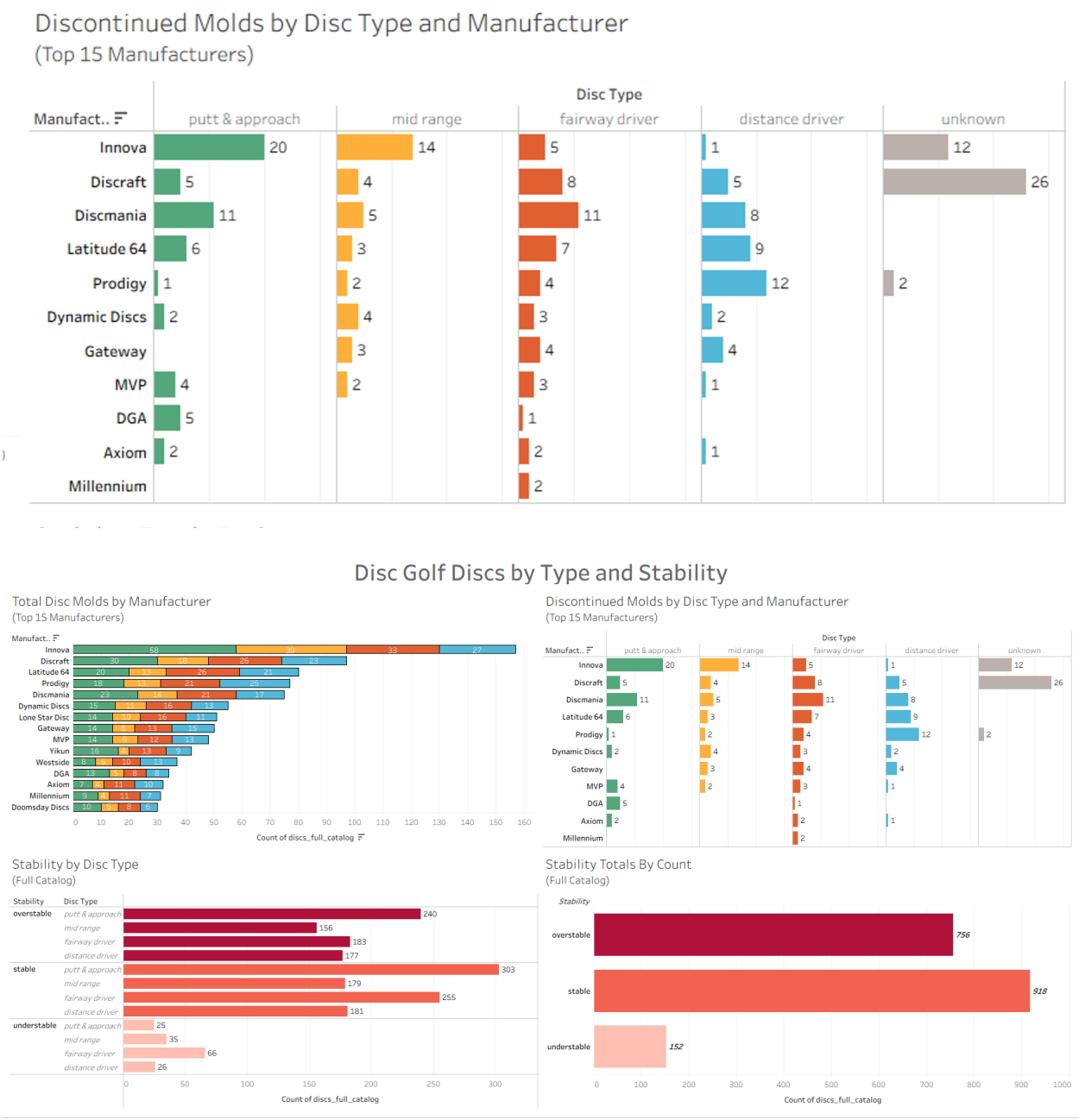


Stability by Disc Type
(Full Catalog)



Stability Totals By Count
(Full Catalog)





Limitations

The limitations of this analysis is that of manufacturers determine their discs flight numbers. Discraft for example had an additional flight number on their discs until 2026. They, Discraft, removed that number to be more inline with other manufacturers. Another limitation is that discs come in different plastics and not every mold behaves the exact same in those plastics. Another limitation is that the data set was compiled via web scrapping and may not be a complete catalog from every manufacturer.

Future Analysis

There are some items that this analysis does not fully cover. Future analysis would include comparing the web scrapped data set with the Professional Disc Golf Associations approved disc list for sanctioned play. This would verify if the data set is a fully complete set that represents all approved game discs. Further analysis would include user flight ratings feedback on the different plastics each disc is released in.

Conclusion

The conclusion of this analysis is that disc type and stability are related but the strength of that relation are unknown and will require further analysis.

Tools

VSCode

Github

Python